

Multiple Choice (10 marks)

1. Statement 1 | If H and K are subgroups of a group G , then $|HK| = |H||K|/|H \cap K|$. Statement 2 | A group of order $2p$ where p is an odd prime is cyclic or isomorphic to D_p .
 - (a) True, False
 - (b) True, True
 - (c) False, False
 - (d) False, True
2. Which statement is true?
 - (a) All trapezoids are rectangles because they have at least one pair of parallel sides.
 - (b) All pentagons are heptagons because they have more than 4 sides.
 - (c) All hexagons are triangles because they have at least 3 sides.
 - (d) All triangles are polygons because they have less than 4 sides.
 - (e) All rectangles are squares because they have 4 right angles.
3. How many ways are there to color the vertices of a cube with two colors, up to rotation?
 - (a) 10
 - (b) 30
 - (c) 23
 - (d) 15
 - (e) 18
4. A model rocket follows the trajectory $c(t) = (80t, 200t - 4.9t^2)$ until it hits the ground, with t in seconds and distance in meters. Find the rocket's maximum height in meters.
 - (a) 2200
 - (b) 1800
 - (c) 2041
 - (d) 2100
 - (e) 1700
5. A zoo has 15 toucans and 60 parrots. What is the ratio of the number of toucans to the number of parrots at the zoo?

- (a) 1:03
- (b) 60:15
- (c) 4:01
- (d) 1:04
- (e) 3:01

True / False (10 marks)

Answer True or False

6. True or False: The magnitude of the vector $\mathbf{v} = (5, 0, 7)$ is 6.0, calculated using the formula $\sqrt{5^2 + 0^2 + 7^2}$.
7. True or False: The upward velocity of the satellite at $t = 1$ is equal to 6.723.
8. True or False: The probability $P(W(1) + W(2) > 2)$ is equal to 0.500, indicating a symmetric distribution around zero.
9. True or False: If a set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly independent, then the dimension of the span V is equal to k .
10. True or False: The correct answer to 'what is the value of $\int_0^\pi (\sin(123x/2)/\sin(x/2))^2 dx$? Round the answer to the thousands decimal.' is '654.3210987'.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. If $Q = 5 + 2i$, $E = i$, and $D = 5 - 2i$, find $Q \cdot E \cdot D$.
12. Express the quadratic function $3x^2 + x - 4$ in the vertex form $a(x - h)^2 + k$. Analyze the process and determine the value of k , providing justification for your steps.

End of Paper